What is science?

Companion text to the lecture Scientific Working, Part 1

Mohcine Chraibi, Armin Seyfried

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This text accompanies the lecture slides. It contains what was said in the lecture but not written on the slides. It follows the lecture's path in five steps:

Problem → Explanation → Testing → Improvement → Application

The guiding question is: how do we gain scientific knowledge, and how do we recognise a good explanation?

A question to start

Why do we know more today than 300 years ago? Obvious answers are experiments, better measuring instruments, more data, better technology. But more observations alone are not enough. We need explanations that we can test critically and improve. This idea comes from David Deutsch (*The Beginning of Infinity*, 2011), who builds on Karl Popper.

An example that explanations do not simply follow from data: Dietrich Braess found the paradox named after him in 1968 in a model, before anyone had observed it on a road.

Step 1: Problem

Why does the earth quake?

The same observation can have very different explanations. An old story, told in many variants, has the earth resting on a bull. The miniature of 1595 from al-Qazwīnī's cosmography shows an angel carrying the earth; the angel stands on a rock, the rock on a bull, the bull on a fish. In other versions the bull carries the earth on its horns, and the earth quakes when it shifts it from one horn to the other. This is not the teaching of a particular culture or religion, but one of many old stories.

Why do we accept plate tectonics today and not the bull? Plate movements can be measured (with GPS, a few centimetres per year). The theory yields testable predictions, for example that earthquakes cluster at plate boundaries.

David Deutsch adds another criterion: a good explanation is *hard to vary* without losing its explanatory power. You can replace the bull with an elephant or a dragon, and the story "explains" the quake just as well. It fixes almost nothing that could be tested. With plate tectonics this does not work: change one part, and the measurements no longer fit.

A game: Braess to try yourself

The network has four nodes: Start, A, B and Goal. Two routes lead from Start to Goal, the upper one via A and the lower one via B. Each route has a variable road whose travel time is n minutes when n drivers use it, and a wide, congestion-free road with fixed travel time C. The travel time of a route is the sum of its roads. In the game, C is the number of participants plus 2; with a fixed value such as 20, the paradox disappears below about 14 people.

Round 1. Everyone secretly chooses “upper” or “lower”. Then, over several rounds, anyone who is dissatisfied may switch. An example with 10 people and C = 12: if 7 drive upper and 3 lower, the upper route takes 7 + 12 = 19 minutes and the lower 12 + 3 = 15 minutes. Those on the upper route switch. At 5 to 5 everyone needs 17 minutes; anyone who switches now needs 18. Nobody can improve alone. This state is called a **Nash equilibrium**, in traffic a **Wardrop equilibrium**. A single secret choice is not yet an equilibrium; the equilibrium is reached only when nobody wants to switch any more.

Round 2. The city builds a shortcut from A to B with travel time zero. Now there are three routes: upper, lower and Start → A → B → Goal. For a split with o drivers upper, u lower and a on the shortcut:

- upper: (o + a) + C
- lower: C + (u + a)
- shortcut: (o + a) + (u + a)

Typically almost everyone takes the shortcut. If all N people drive there, each needs N + 0 + N = 2N minutes, with 10 people 20 instead of 17 before. Whoever switches back to the upper route alone needs N + C minutes, more than 2N. So nobody switches voluntarily, although everyone would be faster without the shortcut. This, too, is a Nash equilibrium, but a worse one. The old 5-to-5 split would be better for everyone, but with the shortcut it is no longer an equilibrium.

The game has a name

The game has exactly the structure of the paper by Dietrich Braess, mathematician and Privatdozent in Münster. It appeared in German in 1968 in the journal *Unternehmensforschung*. Braess did not observe the paradox in traffic; he found it in a model: pure theory, basic research. This fits Deutsch: explanations are not derived from data but conjectured and then criticised. Observations in cities came only afterwards.

The path of the idea: Wardrop formulates the equilibrium principle in 1952, Braess finds the surprising consequence in 1968, Cohen and Horowitz show the same paradox with springs and strings in 1991, the English translation makes it widely accessible in 2005, and today it is an argument in traffic planning: more roads do not automatically help.

Knowledge about the model is not yet knowledge about the world

Everyone chooses the fastest route for themselves, and the result is worse for all. That is what our model predicts. But we only played a small game and assumed that travel time grows with the number of vehicles, that everyone chooses the fastest route and that the new road takes no time. Why should it work the same way in real traffic?

Three levels must be kept apart. The mathematical result is knowledge, but knowledge about the model. Whether real traffic behaves like this is an empirical question. Scientific knowledge means building models, testing predictions and clarifying when a model is useful. What comes out of the game in the

lecture hall is also an observation, while the numbers from the calculation are the theoretical equilibrium. If the two differ, that is already the first lesson.

A good explanation or a plausible story?

Two explanations for the result:

- **A:** “The new road makes everything worse because drivers are selfish.”
- **B:** “With the shortcut, the route over both variable roads is the cheapest for each individual. The load there grows until everyone ends up in a worse equilibrium.”

A sounds plausible but names no mechanism. “Selfish” could explain the opposite as well, and A does not say under which conditions the result occurs. A is our bull: “selfish” could be swapped for anything. B names which roads fill up and why. No part of B can be swapped without the calculation breaking. That is why B allows predictions.

The test: what happens if the shortcut costs 5 minutes? B predicts that the paradox remains but becomes weaker. With 18 people the travel time is then about 34 to 35 instead of 36 minutes, without the shortcut 29. From a shortcut travel time of about N/2 + 2 minutes (11 with 18 people, 9 to 10 with 15 people) nobody takes the shortcut any more, and everything is as before. With A this question cannot even be answered.

Deutsch’s criterion says something about the quality of an explanation. It does not replace testing against experience; it complements it.

A good explanation often reaches beyond the case it was meant for. Related effects exist for springs and strings (Cohen & Horowitz 1991) and in power grids. Whether they occur there must be checked separately in each case.

Step 2: Explanation

Model

Without saying so, in the game we used a model, made a prediction and ran a small experiment. A model is a simplified image of reality. What did we leave out?

- People: not everyone chooses the fastest route.
- Traffic: traffic lights, junctions, queues.
- Roads: lengths, widths, capacities.
- Travel time: why exactly n minutes? (Behind this lies the fundamental diagram.)
- Time: everyone decides at the same time.
- Information: how does everyone know the travel times?

Is the model therefore bad? No. A model does not have to contain all details of reality. It has to capture the essential mechanisms for the question at hand adequately, here the interaction of route choice, traffic load and travel time. With only four nodes and five edges, something surprising can be explained. That is the strength of a model: we simplify reality in order to understand a particular mechanism.

A model should be as simple as possible, but complex enough for the question. This is where **Ockham’s razor** fits: if two models explain the observation equally well, the simpler one is preferable. It is not the number of details that makes a model good, but whether it captures the relevant mechanisms and delivers useful explanations and predictions for its purpose.

An open question remains: how do we know that we left out the right things? This question leads to validation (Step 3).

Assumption, theory, hypothesis

The three terms are often mixed up.

- **Assumption:** "Drivers choose the route that is fastest for them." This is a behavioural assumption. It is set, not tested, and it is not a theory. The model consists of the network and this assumption.
- **Theory:** a coherent, consistent system of statements that follows from it. Here, the equilibrium theory of route choice after Wardrop (1952), first principle: in equilibrium, travel times on all used routes are equal, and no unused route is faster.
- **Hypothesis:** a concrete, testable prediction that follows from the theory and can fail against data. Example: "If road Y in network X is closed, the mean travel time decreases."

The touchstone: which observation would refute the hypothesis? If you cannot say, it is not yet a hypothesis.

Verify or falsify?

Karl Popper (*Logik der Forschung*, 1934): general statements can never be proven conclusively, because one has never seen all cases. Every white swan confirms the statement "All swans are white", but does not prove it. A single black swan refutes it.

Applied to traffic: a hundred cities in which a new road helped do not prove that new roads always help. One Braess case suffices to refute this, but only because the statement is formulated universally ("always"). Against "mostly", a single counterexample does not help.

Popper asks: can the statement fail? Deutsch adds: does it explain why it must be so? We look for explanations that do both.

Back to the earthquake: which observation could show that plate tectonics is wrong? For example, if earthquakes were spread evenly over the earth instead of clustering at plate boundaries, or if GPS measurements showed no movement of the plates. The bull story forbids nothing, so no observation can refute it.

Beware of a terminology trap: in simulation, "verification" means something else, namely checking the program (see below).

Step 3: Testing

Empirics: observations from cities

Three frequently cited cases:

- **Stuttgart 1969:** a new road makes traffic worse. Only when it is closed again does traffic improve.
- **New York 1990:** 42nd Street is closed on Earth Day. Congestion, if anything, decreases.
- **Seoul 2003–2005:** an urban motorway is demolished. First traffic gets slower, then many switch to the subway.

Are these evidence for Braess? They are observations after an intervention, not a controlled experiment. What else changed? In Seoul, trips shifted to the subway over years (Chung et al. 2012). Seoul

is therefore more an example of adaptation than a clean Braess case. In reality many things change at the same time. In an experiment one therefore tries to isolate one cause deliberately.

Experiment: is the new road really to blame?

In the game with 15 people, everyone needed about 24 to 25 minutes without the shortcut and theoretically 30 with it (29 and 36 with 18 people). Can we conclude that the new road is the cause? Perhaps participants decided differently the second time because they already knew the game. Perhaps they were better informed, or understood the rules differently.

A clean comparison consists of two trials: trial A without the shortcut (two routes), trial B with the shortcut (the same people, three routes). The number of people, the travel-time rules, the information and as far as possible all other experimental conditions must be kept constant. Further improvements: vary the order of conditions between groups, run several rounds, compare results.

An experiment deliberately changes one condition and studies its effect.

The controlled game studies the effect of the shortcut within the game model. It is not yet empirical evidence for real traffic networks.

Rapoport and colleagues (2009) ran exactly this experiment in the laboratory. Participants made route choices in an experimental traffic network over many rounds, and their payoff depended on the resulting travel costs. With the additional link they ended up in the worse equilibrium and earned less, even after much practice.

Clever Hans

Even careful experiments can miss something decisive. Around 1900 a horse, “Clever Hans”, answered arithmetic questions by tapping its hoof, before large audiences and mostly correctly. Many experts were convinced; a commission found no trick in 1904.

The owner, Wilhelm von Osten, a mathematics teacher and follower of phrenology, was convinced that animals can think. The psychologist Oskar Pfungst was not satisfied with this explanation. He worked thoroughly with Hans and von Osten and systematically changed one factor at a time: whether the questioner knows the answer, whether Hans can see him (blinkers), how far away he stands. The result:

- If the questioner knows the answer, Hans mostly answers correctly.
- If he does not know it, Hans makes many more errors.
- If Hans cannot see the questioner, he also makes many more errors.

Hans read tiny body signals of the questioner, who unconsciously relaxed as soon as the correct number was reached. Two explanations competed: “Hans can do arithmetic” and “Hans recognises unconscious signals”. Both fit the correct answers. But only the second predicts that Hans fails when nobody within sight knows the answer. Such a discriminating experiment produces knowledge. The second explanation is not better because it sounds simpler, but because it passes the targeted test. Von Osten never accepted the result.

The observation, the correct answers, was real. The explanation was wrong; an influencing factor had been overlooked. An experiment is not automatically a good experiment. As with Braess, a correct observation does not guarantee a correct explanation. That many experts were convinced was no argument: knowledge rests on criticism, not on authority.

This gave rise to the term **Clever Hans effect**: the expectations of observers unconsciously influence the behaviour of those observed. Related are the experimenter effect (Rosenthal) and the interviewer effect in surveys. Today's countermeasures are blinding and control conditions. Science therefore means not only looking for evidence for one's own explanation, but finding out where one might be wrong. This also applies to one's own experimental design.

Simulation

Our network has four nodes; you can compute that in your head. A city network has tens of thousands of links, travel times depend nonlinearly on load, and people decide differently. Then you need numerical methods and software, for example SUMO, MATSim, Vissim or, for pedestrians, JuPedSim.

Is the simulation therefore right?

Suppose the fundamental diagram, the relation between density and flow, is well known from experiments. We derive travel times from it and simulate a traffic network. Is the simulation then right? Not necessarily.

If the software finds the analytically computed equilibrium for our Braess network, this only shows that it computes correctly. That is verification, not validation. Even with correct software and a well-supported fundamental diagram the predictions can be wrong: perhaps people choose their routes differently than assumed, perhaps queues, differences in information or temporal effects are missing.

The misconception is: "The input models are validated and the software passes all tests, so the simulation is realistic." Two questions must be distinguished: did we implement the model correctly? And did we choose the right model in the first place?

Verification and validation

- **Verification:** does the program compute what the model says? Was the model implemented correctly and solved numerically? This is checked with tests and comparison with exact solutions; one looks for errors in the code and in the numerics.
- **Validation:** does the model describe reality well enough for the intended purpose? This is checked by comparison with measured data. A model is always valid for a purpose, never absolutely.

Short formula after Roache: verification means solving the equations right; validation means solving the right equations. A program can be perfectly verified and still not valid. This "verification" is not the same as Popper's.

For pedestrian simulations, RiMEA is relevant (Guideline for Microscopic Evacuation Analysis, version 4.1.1, 2025), with test cases 1–16 in Annex 1. RiMEA calls everything "verification", including the comparison with measured data, which according to AIAA would be called validation. A good exercise: which of these tests are actually validation?

Step 4: Improvement

What moves around what?

Even a wrong or limited model can describe observations quite well. Ptolemy's geocentric model predicted planetary positions usefully for centuries and was refined mathematically again and again.

The heliocentric model of Copernicus (1543) was at first not more accurate in every respect; it still used circular orbits.

Only Kepler's elliptical orbits (*Astronomia nova*, 1609), Galileo's observations with the telescope (1610, for example the phases of Venus) and later Newton's theory of gravitation (1687) produced a much more convincing explanation. The development was not "wrong model versus immediately right model". Knowledge grows by improving and replacing models step by step.

It also follows that a model reproducing the data does not yet prove that its mechanism is right. Ptolemy, too, fitted the data. This raises the question of what really caused a result.

Causality

Causality means: A causes B. In the game the chain is clear: the road does not change the travel time directly; it changes behaviour, and behaviour changes the travel time. This can be shown because we compared with and without the road while everything else stayed the same (controlled comparison).

Seoul is harder. After the urban motorway was demolished, traffic first got slower, but not worse in the long run. Was the demolition the cause, or the switch to the subway, driving behaviour, the economy? Without a comparison this is a correlation in time, not proof of causation. And a falsification in Popper's sense requires a reliable observation. That is exactly what is uncertain in the city cases.

Interaction

The causal chain has a catch: drivers also react to each other. Interaction means that two quantities influence each other. In physics this is Newton's third law (actio = reactio): the earth pulls the moon, and the moon pulls the earth. In the Braess game my route choice changes your travel time, your travel time changes your route choice, and that changes mine again. So one cannot simply say that the road is the cause. The result emerges from the interplay. This holds for almost everything in traffic and pedestrian dynamics.

Step 5: Application

Basic and applied research

Basic research wants to understand, without direct use in view. Applied research wants to solve a problem. The boundary is permeable. Braess' paper was mathematics in 1968; today the argument is part of traffic planning. Peter Grünberg (Forschungszentrum Jülich) discovered giant magnetoresistance, at the same time as Albert Fert; it led to read heads for hard disks, and both received the Nobel Prize in 2007. GPS only works with corrections from the theory of relativity. Applied research often results in standards and guidelines (DIN, FGSV, VDI), which should rest on well-founded knowledge.

Transfer: Le Bon and "mass panic"

Can the same thinking be applied to a completely different theory? Gustave Le Bon (*The Crowd*, 1895) still shapes the image of "mass panic" today. According to him, "the fact that they have been transformed into a crowd puts them in possession of a sort of collective mind" (English translation of 1896).

Two thought experiments: you are a Dortmund fan in the Bayern Munich block, and Dortmund scores in the last minute. What do you feel? Your eleven-year-old daughter goes to a K-pop concert with her grandfather. Does the grandfather get hypnotised?

Questions as in the traffic example: what exactly is explained? Which mechanism lies behind the "collective mind"? Which observation would refute the claim? This tests a theory against experience, as Popper demands. In Deutsch's sense one can ask: the "collective mind" is supposed to explain aggressive as well as peaceful crowds. Can it be adapted so freely that it explains everything and therefore nothing?

And your own work?

Every thesis can be placed: is it empirical, theoretical (with a model) or based on simulation? Is it closer to basic research (like Braess 1968) or to application (like traffic planning with simulation)? And the guiding question for every claim: what would be convincing evidence?

Outlook: a hypothesis of your own

"A bottleneck 3 m wide and 1 m long, or 1 m wide and 3 m long: which is better?" As it stands, this is not yet a scientific question. One must fix the walking direction (what counts is the width across the walking direction, not the orientation on paper), the geometry, and what "better" means: higher flow, shorter evacuation time or lower density in front of the bottleneck? Only then does an intuitive question become a testable hypothesis. As with Braess, one changes the geometry and asks what happens to the flow.

What have we learned?

1. We want to understand the world.
2. We discover something surprising.
3. We look for an explanation.
4. We test whether our explanation holds.
5. We improve our knowledge and apply it.

Scientific knowledge is provisional, testable, methodically grounded and correctable. The answer to the opening question, why we know more today than 300 years ago, is therefore: we form explanations, criticise and improve them, instead of merely collecting observations. Examples are the bull and plate tectonics, Ptolemy versus Kepler and Newton (Newton's *Principia* appeared in 1687, a little more than 300 years ago).

> Science does not mean possessing final answers. It means developing explanations, discovering errors and thereby finding ever better explanations.

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